Project Proposal

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1. Proposed Project Title

KodeX

KodeX is a competitive programming platform enabling **problem setters to create problems with hidden and visible test cases**, and **users to solve problems with server-side code execution via Judge0 API**. Both user solution and tester's expected solution are executed against all test cases, automatically compared for exact output matching, and verdicts (AC/WA/TLE/MLE) displayed with detailed per-test-case breakdown. Built with MERN stack, it supports secure code execution, leaderboards.

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2. Team Members and IDs

Sr. No	Name	ID
1	Jeet Maheshwari	202512052
2	Dhumil Kachhiya	202512056
3	Nishit Gal	202512002
4	Ayush Chakarani	202512084

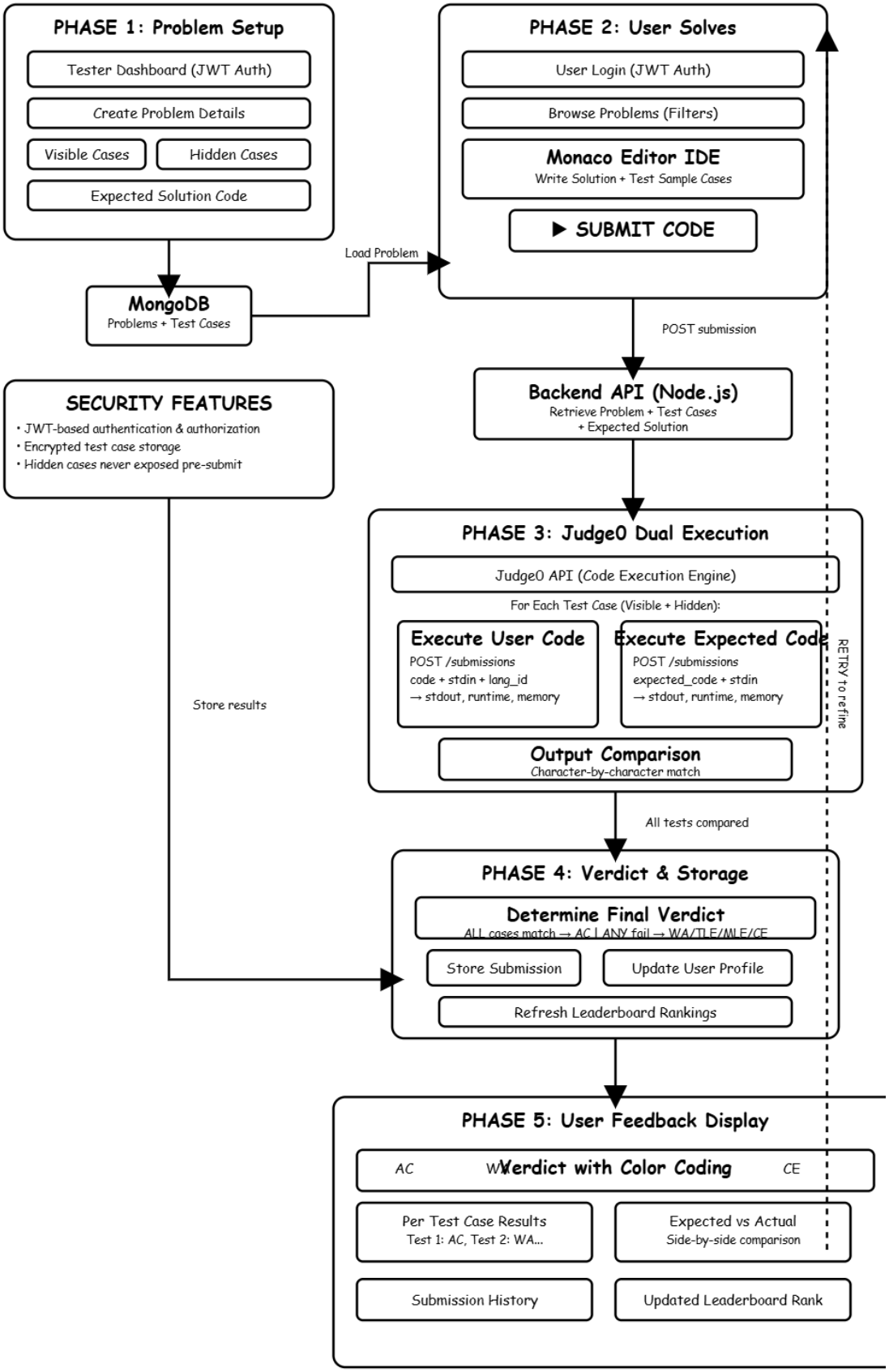
3. List of Functionalities

Core Features:

1. **Problem Setter Dashboard** – Testers create problems with title, difficulty, tags, description, sample test cases (visible), hidden test cases (input/output pairs), and expected solution code. Form validation ensures test case integrity before publication.
2. **User Authentication & Authorization** – JWT-based secure login/signup with role-based access control (student user, problem setter/tester, admin). Passwords hashed with bcrypt.
3. **Problem Repository Browser** – Users browse problems filtered by difficulty (Easy/Medium/Hard), tags (Arrays, Trees, DP, Graphs), and company-specific curated lists. Sample test cases displayed; hidden cases remain concealed.
4. **Online IDE with Monaco Editor** – Browser-based code editor supporting JavaScript, Java, Python, C++ with syntax highlighting, auto-indentation, real-time formatting, and code templates.
5. **Dual Code Execution Engine** – Backend simultaneously executes user solution and tester's expected solution against all test cases (visible + hidden) via Judge0 API. Outputs compared character-by-character for exact matches.
6. **Automated Verdict System** – Display verdicts (AC/WA/TLE/MLE/CE) with detailed metrics: runtime, memory usage, CPU time. Show per-test-case breakdown indicating which cases passed/failed and why (WA → show user vs expected output).
7. **Submission History & Analytics** – Track all submission attempts with code, timestamp, verdict, language, runtime/memory. Users view acceptance rate, total attempts per problem, and time-series of improvement.
8. **Global & Friend Leaderboards** – Real-time rankings by problems solved, acceptance rate, contest performance. Points system: easy +1, medium +2, hard +5. Weekly reset for competitive streaks.

9. **Community Discussion Forum** – Threaded comments per problem for solution hints, approach discussion, and Q&A. Users upvote helpful comments; tester can pin official solutions.
10. **Admin Dashboard** – Add/edit/delete problems, manage test cases, monitor platform analytics (submissions/day, average verdict time), moderate discussions, view server logs, manage user roles.

4. Block Diagram


PHASE 1: Problem Setup
Tester Dashboard (JWT Auth)
Create Problem Details
Visible Cases
Hidden Cases
Expected Solution Code
MongoDB
Problems + Test Cases
Load Problem
PHASE 2: User Solves
User Login (JWT Auth)
Browse Problems (Filters)
Monaco Editor IDE
Write Solution + Test Sample Cases
▶ SUBMIT CODE
POST submission
Backend API (Node.js)
Retrieve Problem + Test Cases
+ Expected Solution
SECURITY FEATURES
• JWT-based authentication & authorization
• Encrypted test case storage
• Hidden cases never exposed pre-submit
Store results
PHASE 3: Judge0 Dual Execution
Judge0 API (Code Execution Engine)
For Each Test Case (Visible + Hidden):
Execute User Code
POST /submissions
code + stdin + lang_id
→ stdout, runtime, memory
Execute Expected Code
POST /submissions
expected_code + stdin
→ stdout, runtime, memory
Output Comparison
Character-by-character match
RETRY to refine
All tests compared
PHASE 4: Verdict & Storage
Determine Final Verdict
ALL cases match → AC | ANY fail → WA/TLE/MLE/CE
Store Submission
Update User Profile
Refresh Leaderboard Rankings
PHASE 5: User Feedback Display
AC
Verdict with Color Coding
CE
Per Test Case Results
Test 1: AC, Test 2: WA...
Expected vs Actual
Side-by-side comparison
Submission History
Updated Leaderboard Rank

5. Tentative Work Distribution

Name	Role
Jeet Maheshwari	Full stack
Dhumil Kachhiya	Backend
Nishit Gal	Backend
Ayush Chakarani	Frontend

6. Technologies to be Used

Frontend Technologies

- React.js 18+
- Tailwind CSS
- Monaco Editor
- Axios
- Redux Toolkit
- React Router v6

Backend Technologies

- Node.js 18+
- Express.js 4
- Judge0 CE API
- JWT (jsonwebtoken)
- Bcrypt
- Axios
- Dotenv
- CORS
- Helmet

Database & Infrastructure

- MongoDB
- Mongoose 7+
- MongoDB Atlas
- GitHub

Development & Testing Tools

- VS Code
- Postman / Thunder Client
- Git + GitHub
- ESLint + Prettier